

Question	Answer	Marks
1(a)	mass per unit volume	B1
1(b)(i)	density = $0.243 / (0.0541 \times 0.1109 \times 0.0162)$	C1
	= 2500 kg m^{-3}	A1
1(b)(ii)	($0.001 / 0.243$) or ($0.01 / 5.41$) or ($0.01 / 11.09$) or ($0.01 / 1.62$)	C1
	$(\Delta\rho / \rho) = (\Delta m / m) + (\Delta x / x) + (\Delta y / y) + (\Delta z / z)$ $= (0.001 / 0.243) + (0.01 / 5.41) + (0.01 / 11.09) + (0.01 / 1.62)$ $(= 0.013)$	C1
	percentage uncertainty in $\rho = 0.013 \times 100$ $= 1.3\%$ (allow 1 s.f. answer of 1%)	A1
1(c)	zero error (on calipers / balance) or incorrect calibration (of calipers / balance)	B1

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